# Rajalakshmi Engineering College

## FUNDAMENTALS OF DATA SCIENCE LAB

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## EXPERIMENT 13

Handle missing values and inappropriate data

# Experiment 13: Handle missing values and inappropriate data

## Aim

To handle missing values and inappropriate data using pandas.

## Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv("Hotel_Dataset.csv")
df
df.duplicated()
df.info()
df.drop_duplicates(inplace=True)
df
len(df)
index=np.array(list(range(0,len(df))))
df.set_index(index,inplace=True)
index
df
df.drop(['Age_Group.1'],axis=1,inplace=True)
df
df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
df
df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
df
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
df.Age_Group.unique()
df.Hotel.unique()
df.Hotel.replace(['Ibys'],'Ibis',inplace=True)
df.FoodPreference.unique
df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)
df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)
df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
df.Bill.fillna(round(df.Bill.mean()),inplace=True)
df
```

## Output - Experiment 13

| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | EstimatedSalary | Age_Group.1 |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 20-25 | 4 | Ibis | veg | 1300 | 2 | 40000 | 20-25 |
| **1** | 2 | 30-35 | 5 | LemonTree | Non-Veg | 2000 | 3 | 59000 | 30-35 |
| **2** | 3 | 25-30 | 6 | RedFox | Veg | 1322 | 2 | 30000 | 25-30 |
| **3** | 4 | 20-25 | -1 | LemonTree | Veg | 1234 | 2 | 120000 | 20-25 |
| **4** | 5 | 35+ | 3 | Ibis | Vegetarian | 989 | 2 | 45000 | 35+ |
| **5** | 6 | 35+ | 3 | Ibys | Non-Veg | 1909 | 2 | 122220 | 35+ |
| **6** | 7 | 35+ | 4 | RedFox | Vegetarian | 1000 | -1 | 21122 | 35+ |
| **7** | 8 | 20-25 | 7 | LemonTree | Veg | 2999 | -10 | 345673 | 20-25 |
| **8** | 9 | 25-30 | 2 | Ibis | Non-Veg | 3456 | 3 | -99999 | 25-30 |
| **9** | 9 | 25-30 | 2 | Ibis | Non-Veg | 3456 | 3 | -99999 | 25-30 |
| **10** | 10 | 30-35 | 5 | RedFox | non-Veg | -6755 | 4 | 87777 | 30-35 |

**Output - Experiment 13**

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

**Output - Experiment 13**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        11 non-null     int64
 1   Age_Group         11 non-null     object
 2   Rating(1-5)       11 non-null     int64
 3   Hotel             11 non-null     object
 4   FoodPreference    11 non-null     object
 5   Bill              11 non-null     int64
 6   NoOfPax           11 non-null     int64
 7   EstimatedSalary   11 non-null     int64
 8   Age_Group.1       11 non-null     object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

## Output - Experiment 13

| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | EstimatedSalary | Age_Group.1 |
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| **3** | 4 | 20-25 | -1 | LemonTree | Veg | 1234 | 2 | 120000 | 20-25 |
| **4** | 5 | 35+ | 3 | Ibis | Vegetarian | 989 | 2 | 45000 | 35+ |
| **5** | 6 | 35+ | 3 | Ibys | Non-Veg | 1909 | 2 | 122220 | 35+ |
| **6** | 7 | 35+ | 4 | RedFox | Vegetarian | 1000 | -1 | 21122 | 35+ |
| **7** | 8 | 20-25 | 7 | LemonTree | Veg | 2999 | -10 | 345673 | 20-25 |
| **8** | 9 | 25-30 | 2 | Ibis | Non-Veg | 3456 | 3 | -99999 | 25-30 |
| **10** | 10 | 30-35 | 5 | RedFox | non-Veg | -6755 | 4 | 87777 | 30-35 |

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| 2 | 3 | 25-30 | 6 | RedFox | Veg | 1322 | 2 | 30000 | 25-30 |
| 3 | 4 | 20-25 | -1 | LemonTree | Veg | 1234 | 2 | 120000 | 20-25 |
| 4 | 5 | 35+ | 3 | Ibis | Vegetarian | 989 | 2 | 45000 | 35+ |
| 5 | 6 | 35+ | 3 | Ibys | Non-Veg | 1909 | 2 | 122220 | 35+ |
| 6 | 7 | 35+ | 4 | RedFox | Vegetarian | 1000 | -1 | 21122 | 35+ |
| 7 | 8 | 20-25 | 7 | LemonTree | Veg | 2999 | -10 | 345673 | 20-25 |
| 8 | 9 | 25-30 | 2 | Ibis | Non-Veg | 3456 | 3 | -99999 | 25-30 |
| 9 | 10 | 30-35 | 5 | RedFox | non-Veg | -6755 | 4 | 87777 | 30-35 |

## Output - Experiment 13

| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | EstimatedSalary |
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| 0 | 1 | 20-25 | 4 | Ibis | veg | 1300 | 2 | 40000 |
| 1 | 2 | 30-35 | 5 | LemonTree | Non-Veg | 2000 | 3 | 59000 |
| 2 | 3 | 25-30 | 6 | RedFox | Veg | 1322 | 2 | 30000 |
| 3 | 4 | 20-25 | -1 | LemonTree | Veg | 1234 | 2 | 120000 |
| 4 | 5 | 35+ | 3 | Ibis | Vegetarian | 989 | 2 | 45000 |
| 5 | 6 | 35+ | 3 | Ibys | Non-Veg | 1909 | 2 | 122220 |
| 6 | 7 | 35+ | 4 | RedFox | Vegetarian | 1000 | -1 | 21122 |
| 7 | 8 | 20-25 | 7 | LemonTree | Veg | 2999 | -10 | 345673 |
| 8 | 9 | 25-30 | 2 | Ibis | Non-Veg | 3456 | 3 | -99999 |
| 9 | 10 | 30-35 | 5 | RedFox | non-Veg | -6755 | 4 | 87777 |

## Output - Experiment 13

| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | Estimated Salary |
|---|---|---|---|---|---|---|---|---|
| 0 | 1.0 | 20-25 | 4.0 | Ibis | veg | 1300.0 | 2 | 40000.0 |
| 1 | 2.0 | 30-35 | 5.0 | LemonTree | Non-Veg | 2000.0 | 3 | 59000.0 |
| 2 | 3.0 | 25-30 | NaN | RedFox | Veg | 1322.0 | 2 | 30000.0 |
| 3 | 4.0 | 20-25 | NaN | LemonTree | Veg | 1234.0 | 2 | 120000.0 |
| 4 | 5.0 | 35+ | 3.0 | Ibis | Vegetarian | 989.0 | 2 | 45000.0 |
| 5 | 6.0 | 35+ | 3.0 | Ibys | Non-Veg | 1909.0 | 2 | 122220.0 |
| 6 | 7.0 | 35+ | 4.0 | RedFox | Vegetarian | 1000.0 | -1 | 21122.0 |
| 7 | 8.0 | 20-25 | NaN | LemonTree | Veg | 2999.0 | -10 | 345673.0 |
| 8 | 9.0 | 25-30 | 2.0 | Ibis | Non-Veg | 3456.0 | 3 | NaN |
| 9 | 10.0 | 30-35 | 5.0 | RedFox | non-Veg | NaN | 4 | 87777.0 |

## Output - Experiment 13

| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | EstimatedSalary |
|---|---|---|---|---|---|---|---|---|
| **0** | 1.0 | 20-25 | 4.0 | Ibis | veg | 1300.0 | 2.0 | 40000.0 |
| **1** | 2.0 | 30-35 | 5.0 | LemonTree | Non-Veg | 2000.0 | 3.0 | 59000.0 |
| **2** | 3.0 | 25-30 | NaN | RedFox | Veg | 1322.0 | 2.0 | 30000.0 |
| **3** | 4.0 | 20-25 | NaN | LemonTree | Veg | 1234.0 | 2.0 | 120000.0 |
| **4** | 5.0 | 35+ | 3.0 | Ibis | Vegetarian | 989.0 | 2.0 | 45000.0 |
| **5** | 6.0 | 35+ | 3.0 | Ibys | Non-Veg | 1909.0 | 2.0 | 122220.0 |
| **6** | 7.0 | 35+ | 4.0 | RedFox | Vegetarian | 1000.0 | NaN | 21122.0 |
| **7** | 8.0 | 20-25 | NaN | LemonTree | Veg | 2999.0 | NaN | 345673.0 |
| **8** | 9.0 | 25-30 | 2.0 | Ibis | Non-Veg | 3456.0 | 3.0 | NaN |
| **9** | 10.0 | 30-35 | 5.0 | RedFox | non-Veg | NaN | 4.0 | 87777.0 |

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| | CustomerID | Age_Group | Rating(1-5) | Hotel | FoodPreference | Bill | NoOfPax | EstimatedSalary |
|---|---|---|---|---|---|---|---|---|
| 0 | 1.0 | 20-25 | 4.0 | Ibis | Veg | 1300.0 | 2.0 | 40000.0 |
| 1 | 2.0 | 30-35 | 5.0 | LemonTree | Non-Veg | 2000.0 | 3.0 | 59000.0 |
| 2 | 3.0 | 25-30 | 4.0 | RedFox | Veg | 1322.0 | 2.0 | 30000.0 |
| 3 | 4.0 | 20-25 | 4.0 | LemonTree | Veg | 1234.0 | 2.0 | 120000.0 |
| 4 | 5.0 | 35+ | 3.0 | Ibis | Veg | 989.0 | 2.0 | 45000.0 |
| 5 | 6.0 | 35+ | 3.0 | Ibis | Non-Veg | 1909.0 | 2.0 | 122220.0 |
| 6 | 7.0 | 35+ | 4.0 | RedFox | Veg | 1000.0 | 2.0 | 21122.0 |
| 7 | 8.0 | 20-25 | 4.0 | LemonTree | Veg | 2999.0 | 2.0 | 345673.0 |
| 8 | 9.0 | 25-30 | 2.0 | Ibis | Non-Veg | 3456.0 | 3.0 | 96755.0 |
| 9 | 10.0 | 30-35 | 5.0 | RedFox | Non-Veg | 1801.0 | 4.0 | 87777.0 |

## Result - Experiment 13

Missing values and inappropriate data in the dataset were handled.